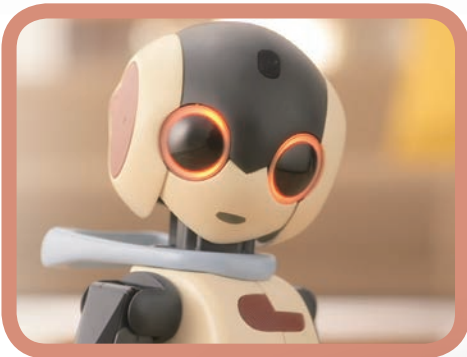
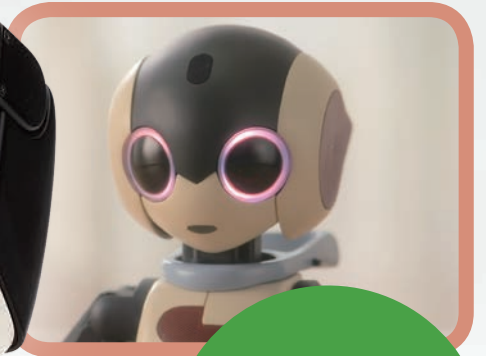


Build your own **Robi**

Pack 10



Stages 35-38:
Begin building up
Robi's left leg

Build your own **Robi**

CONTENTS

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Assembly Guide

p175

Stage 35: Prepare Robi's left foot servo p175

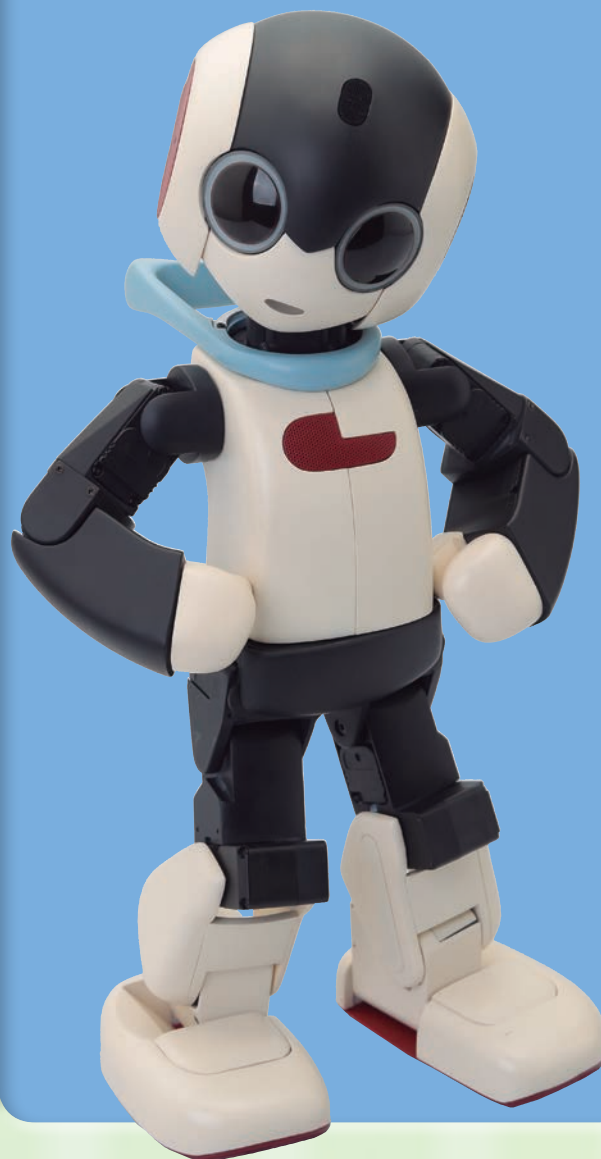
Stage 36: Fitting the left foot servo cover p179

Stage 37: Joining the left foot servo to the ankle frame p182

Stage 38: Robi's battery pack p186

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THIS PRODUCT IS NOT A TOY AND IS NOT DESIGNED FOR USE IN
PLAY. ITEMS MAY VARY FROM THOSE SHOWN. THERE MAY BE
COLOUR VARIATIONS OWING TO THE MATERIALS USED.



STAGE 35: PREPARE ROBI'S LEFT FOOT SERVO

Now that you have built up Robi's right leg from foot to hip, it is time to repeat this process for his left leg.

In this session, you will begin building up Robi's left leg, which will be done in much the same way as you did his right. Again, you will start from the foot, then work up through his ankle and lower leg frames, before finishing

the leg by adding the knee and hip joints. However, you must still follow the steps closely, as going back to re-wire a servo cable, or re-program an already fitted servo, can be a lengthy process!



1



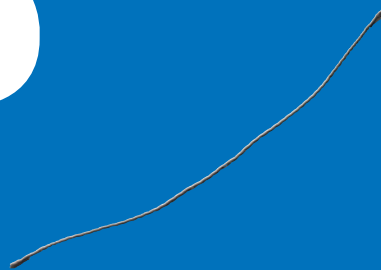
YOUR PARTS

1 Servo motor (ID=11)

**PARTS TO BE
ASSEMBLED**

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



*Servo cable
(Stage 34)*



*Neck stand
(Stage 8)*

THE SERVO AND CABLE

TIP!



As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure you hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

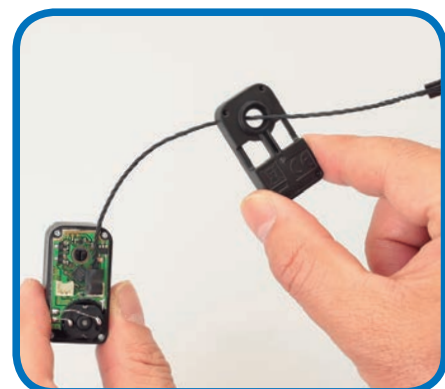
2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath it.



If you find it difficult to attach the cable with the pad in place, or if you chose not to fit it with pads in the previous stage, it is possible to apply the pad after fitting the connector. Take extra care when handling the unprotected

connector. Also be careful not to press down on or obstruct the circled components on the servo's circuit board.

3 Take the long servo cable prepared in Stage 34 and fit one of its connectors onto one of the two sockets, as shown (it does not matter which one). As ever, take great care when handling exposed electronic components.

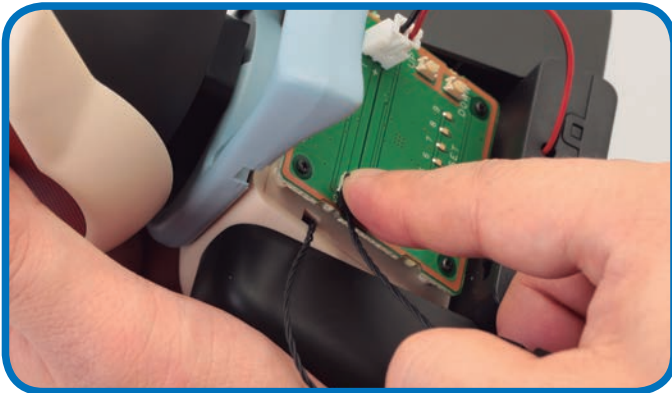


4 Take the servo cover removed in Step 1 and fit it back onto the open servo, feeding the free tip of the servo cable through its central hole as you do.

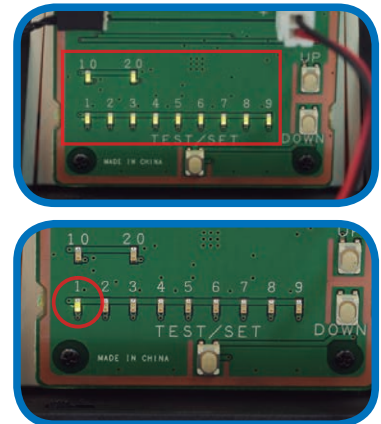
5 Press the cover securely back into place.

6 Lightly tighten all four screws removed in Step 1 into each corner of the servo cover.

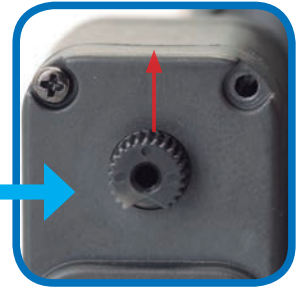
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board, and attach the servo cable connected to the servo in Step 3.

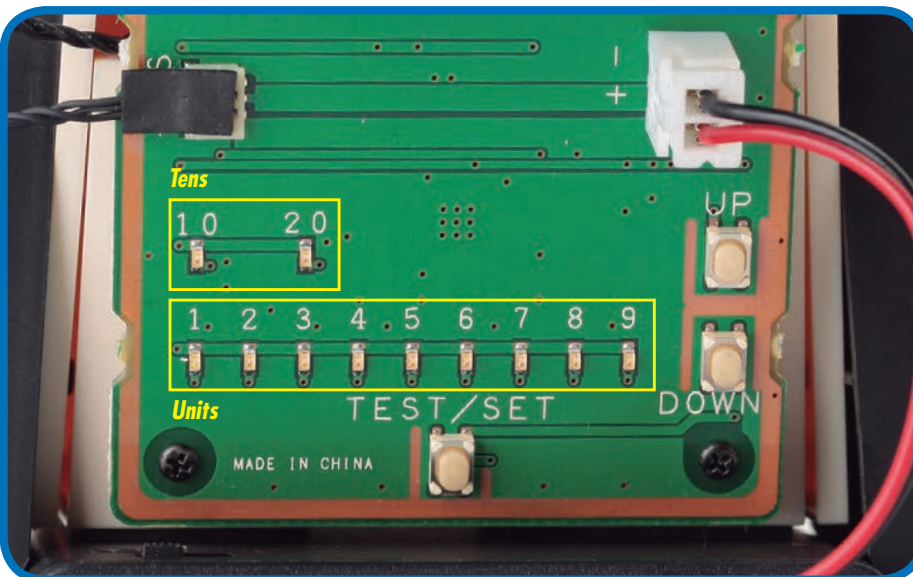


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

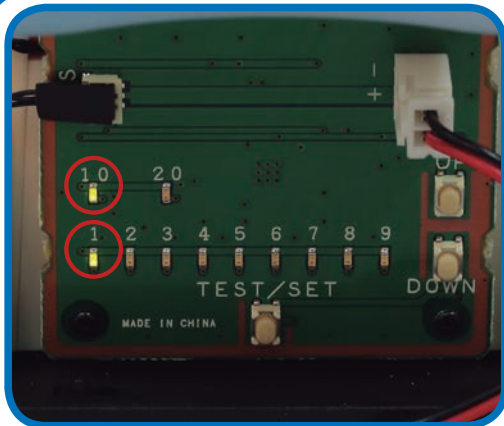


- 9** To test the servo, press the TEST/SET switch: the servo shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

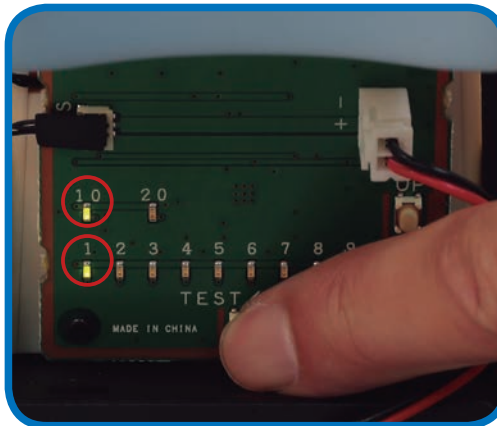
SETTING THE ID NUMBER



- 10** Re-familiarise yourself with the lights and switches on the test board. These are used to set the ID numbers of the servos. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 11, as you will do in this stage, you press UP 10 times. This will light up the 10 and 1 LEDs. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, set it by pressing and holding the TEST/SET switch.



- 11** To set the ID number of Robi's left foot servo to 11, press the UP button 10 times. Make sure that both the 10 and 1 LED is lit, using the UP and DOWN buttons to correct the number if necessary.

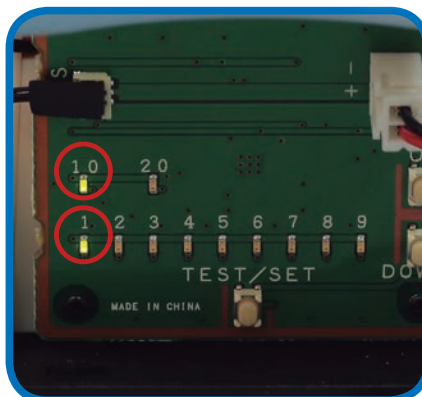
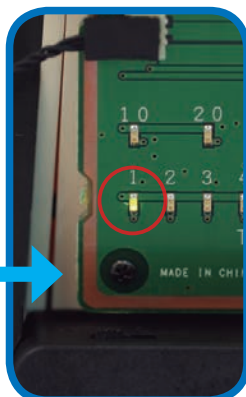
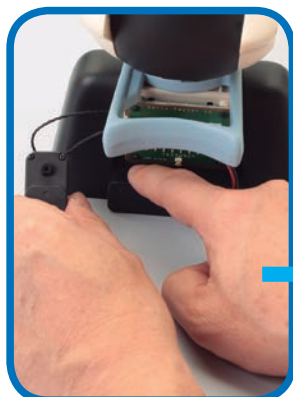


- 12** Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 10 and 1 LEDs should flash a few times, then remain lit.



ID=11

CHECKING THE ID NUMBER



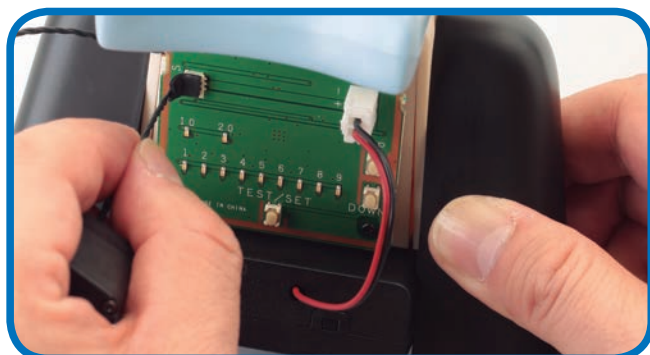
HELP!



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.

- 13** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.

- 14** To check the ID number, press TEST/SET. Now the 10 and 1 LEDs should light up and the servo shaft should repeat the rotations seen in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.



- 15** Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit into Robi's leg.



Prepared left foot servo

The servo that will tilt Robi's left foot is assembled, wired and set. Make sure you keep this part safely, preferably in a sealed plastic bag with its ID number (11) written on it for ease of reference.

STAGE 36: FITTING THE LEFT FOOT SERVO COVER

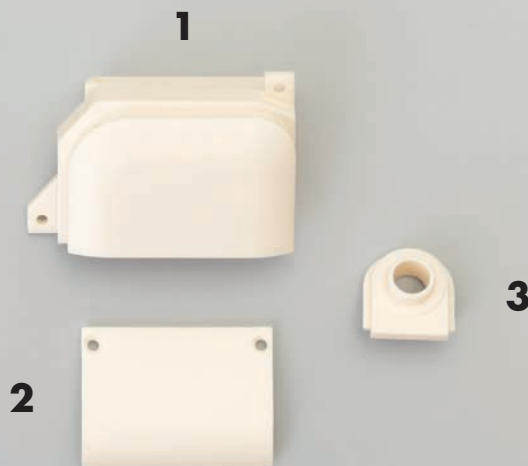
In this assembly you will fit the servo prepared in Stage 35 (ID=11) to its cover, as Robi's left foot starts to take shape.

The servo supplied and prepared in the previous stage, ID=11, will tilt Robi's left foot from side to side on completion. In this stage you will fit it to its cover, which is coloured and moulded to fit the rest of Robi's

exoskeleton. Along with the cover that you will use in this build, you have received other parts that will make up Robi's left ankle – you will not be using these now, so be sure to keep them safely until they are needed.



**PARTS TO BE
ASSEMBLED**



YOUR PARTS

- 1 Left foot cover
- 2 Left ankle side panel
- 3 Left ankle joint

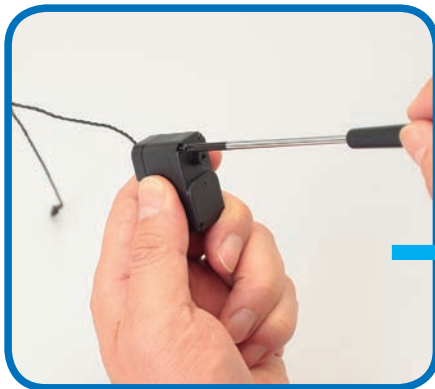
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

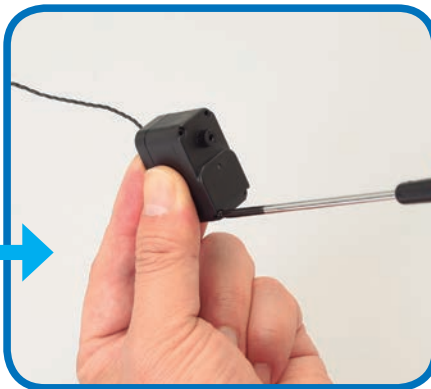


Wired servo (Stage 35)

PREPARE THE SERVO

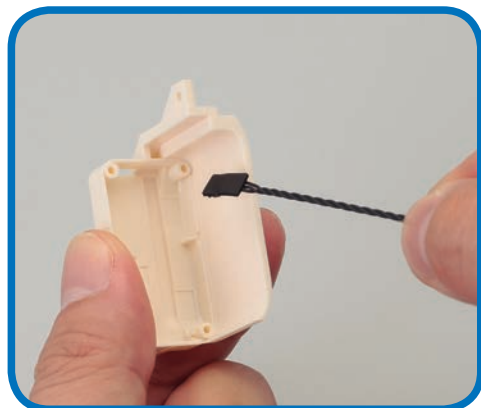


- 1** Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

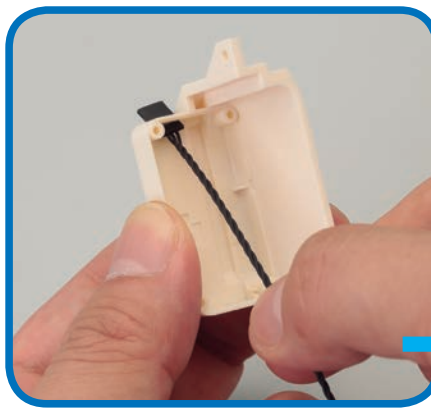


- 2** Gently lift the cover away, taking care not to force it or touch the circuit board underneath.

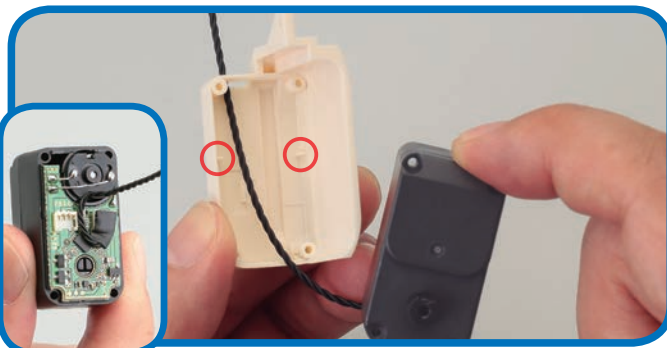
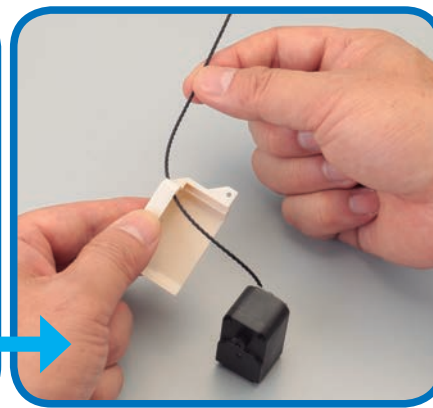
THE LEFT FOOT COVER



- 3** Line up the free end of the prepared servo's cable to the left foot cover, as shown.



- 4** Carefully feed the servo's connector through the rectangular hole in the cover, as shown. Do not pull too hard as you do this, and do not let the cable get caught. Stop when about two thirds of the cable is through.



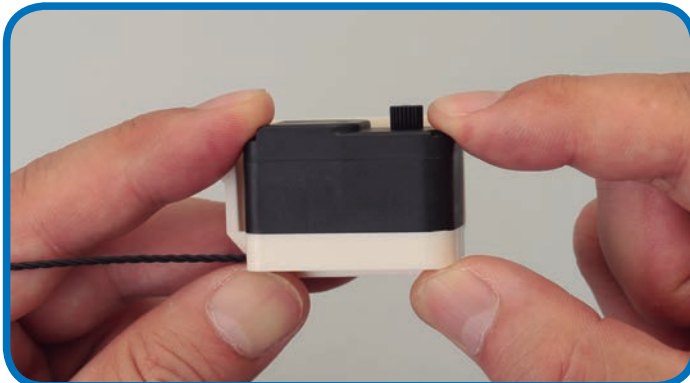
- 5** Lower the open-faced servo into the cover, feeding the cable through the hole as you do so. Be very careful that the cable does not bunch up inside the cover, and that it does not get caught on the two circled projections, as this may damage it. Turning the cable neatly over on itself before lowering the servo into place (inset) will help avoid this.



- 6** With the cable neatly folded and leading through the hole as shown, lower the servo fully to the cover. Keep the cable slightly taut as you do this, to avoid any slack forming inside the cover.



7 Press the servo carefully into place.



8 Check that the servo and cover are meeting neatly along their edges, and that no gaps or evidence of the cable having been trapped or pinched can be seen.



9 Use the four screws removed in Step 1 to fasten the servo to its cover.



TIP!

Assembled foot servo

The servo that will tilt Robi's left foot from side to side is wired and fitted to its cover. Make sure you keep this subassembly safely, ideally in a sealed plastic bag.



STORAGE OF PARTS



In this stage, you have received parts that are not used in this assembly's steps. These should be stored neatly away, ideally in a compartmentalised hobby box, as mentioned previously. We also remind you that it is advisable that you keep all Robi's as-yet unused plastic parts out of direct sunlight, as well as separated by colour, to preserve their appearance until they are needed.

STAGE 37: JOINING THE LEFT FOOT SERVO TO THE ANKLE FRAME

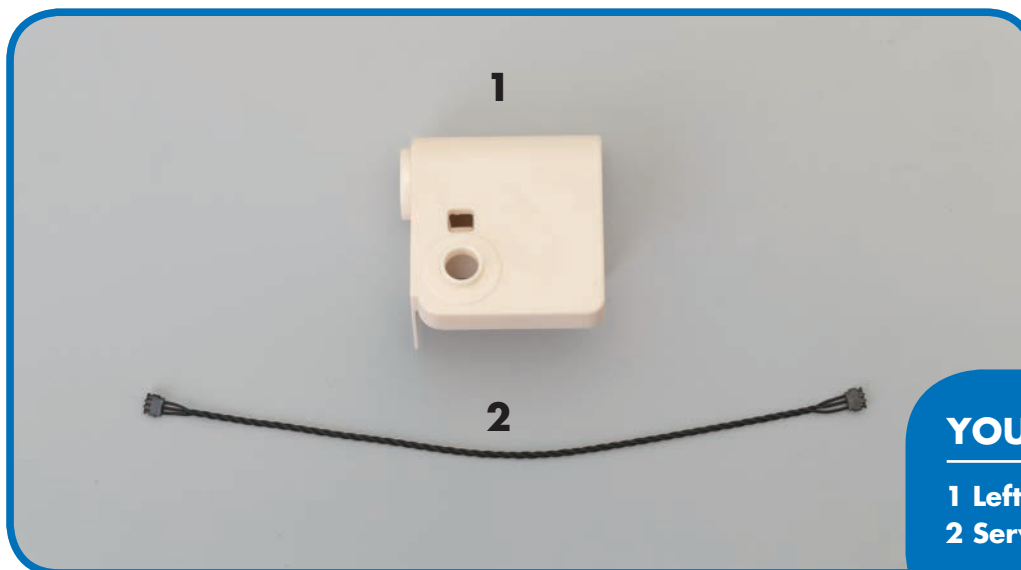
In this assembly, you will join the left foot cover and servo subassembly built in the previous stage to its ankle frame, ready to connect the ankle servo in the coming stages.

Robi's foot servo will work in tandem with his ankle servo, the side-to-side tilting motion of the former working with the up-and-down rotation of the latter, giving him the flexibility needed to walk naturally. In this stage you prepare to connect these two servos,

by joining the frame that will hold Robi's ankle servo to the shaft of the left foot servo (ID = 11) from the previous stage. The work mirrors that of Stage 25, where you did the same for the right foot, but as always take care to follow these steps closely.



PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Left ankle frame
- 2 Servo cable (135mm)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

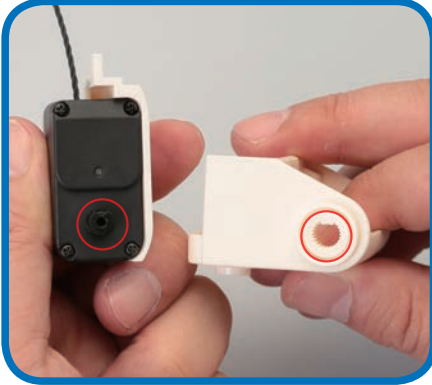


Servo and foot cover subassembly (Stage 36)

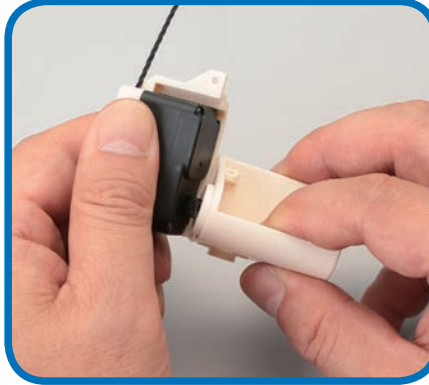


Protective pads (Stage 3)

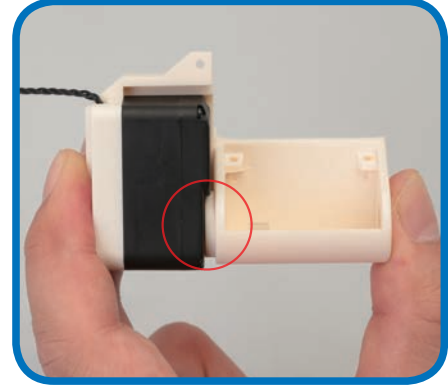
JOINING THE FOOT SERVO TO THE ANKLE FRAME



- 1 With the foot servo and ankle frame positioned at right angles as shown, line up the servo's shaft to the hole in the ankle frame. Note the location of the notch in the servo's shaft, which should line up exactly to the notch in the frame's D-cut hole.



- 2 With the parts lined up exactly, press the ankle frame onto the servo's shaft, making absolutely sure not to force the parts (see the Close-Up box below).



- 3 Check that the parts are joined neatly, and that no gap can be seen at the point where the shaft enters the frame. The parts should be positioned at right angles to one another.

HANDLE WITH CARE!



This stage sees you working with D-cuts once again. As with the previous instances of these, care must be taken to ensure that the correspondingly shaped parts are fitted together properly, as this will ensure the correct 'neutral' position is achieved, and avoid irreversible damage.

Locate the D-cuts

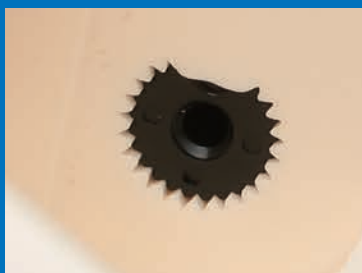
The servo shaft is not fully cylindrical, but has a small, triangular notch in it. This creates the outline of a letter 'D', as shown in the photo near right. To match this, the corresponding hole in the ankle frame has a small flat section in it, again creating the outline of a letter 'D' (far right).



Along with the smaller teeth used to grip each other, the parts have corresponding notches and indents designed to ensure the correct neutral position is achieved.

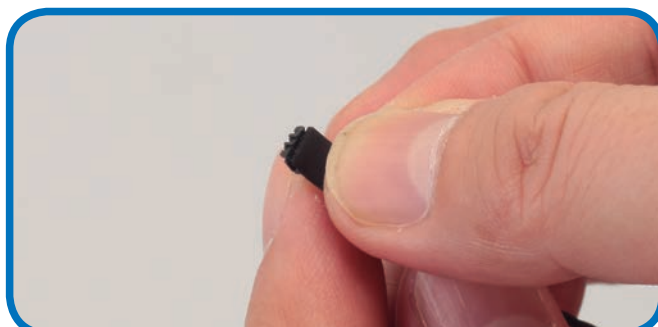
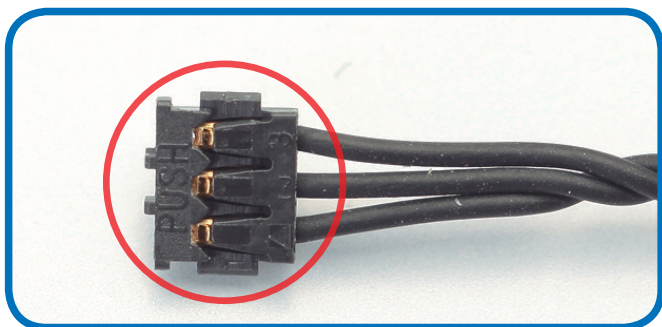
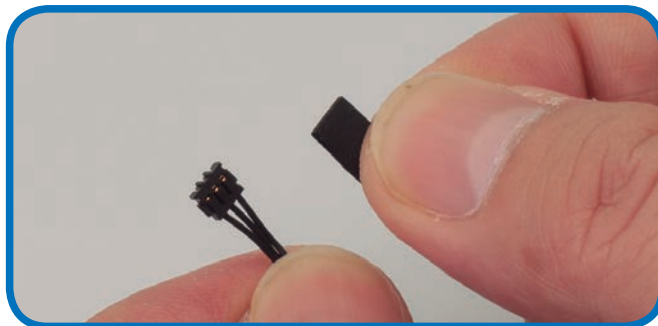
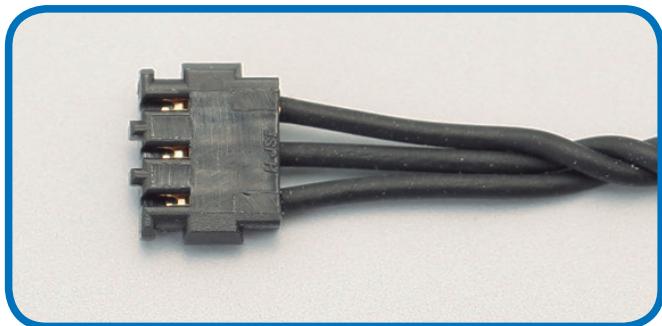
A clean fit

The frame's hole is designed to match the servo shaft, with the D-cut outline of the shaft fitting to the frame exactly. It is vital that the parts connect neatly, as severe damage can occur otherwise.



The frame should fit over the servo's shaft neatly and easily. Never force it, as this may well result in irreparable damage to the frame, as seen in the photo near left.

SEALING THE SERVO CABLE



- 4** As you have done previously, identify the side of this stage's servo cable's connector that has PUSH marked on it (circled).

- 5** Apply a protective pad to the connector on this side. Repeat for both ends of the cable.

TIP!

During the build so far, you may have found it easier to apply the protective pads after fitting the cables to the servos' sockets. If this is the case, you can leave the cable's connectors uncovered for now, and fit the pads later.

Assembled foot servo and ankle frame

The servo that will tilt Robi's left foot from side to side is now joined to the frame that will hold Robi's ankle servo. Make sure you keep this subassembly safely until it is next needed.



USEFUL TOOLS AND TIPS



In this section we look at how to work with and identify the different screws used throughout the build, as well as some tips on using batteries.

M2 x 6mm countersunk screw

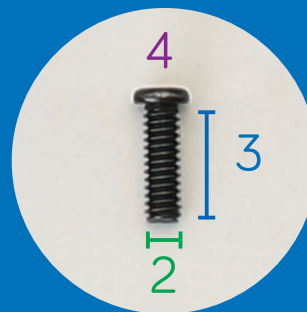
1 2 3 4

1. Specification: indicates that this screw has a metric thread, measured in millimetres.
2. Diameter: the 'thickness' of the screw, in the case of the screw above, 2mm.
3. Length: the length of the threaded section that will sit within the hole – however, in the case of countersunk screws, where the shaped head also sits within the hole, it is the length of the entire screw that this represents (see right).
4. Type: a countersunk screw has a flat head that sits within the hole, a button- or pan-head one has a raised, rounded head; while a self-tapping screw has wider and more sloped threads designed to bite into an un-tapped hole (one without pre-cut grooves for the screw to follow). As self-tapping and countersunk varieties are more specialised, where a screw is listed simply as a 'screw', it will be a non-self-tapping, pan/button-head one.

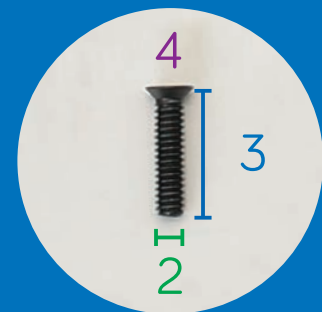
Screw types

A few variations of screws have been used in the assembly stages so far, for example the countersunk ones that have a flat head designed to sit flush with a surface when fitted. All, though, are 'metric' screws, meaning they have been manufactured and measured in line with the ISO universal metric standard, which is noted by the 'M' at the start of their names.

Some of the screws may be listed as being the same 'size', for example 'M2 x 6mm', but the type of screw, given after the measurement, means these are not always identical, so it is a good idea to familiarise yourself with these universal classifications.



Pan-head screw



Countersunk screw

Magnetic screw dish

The screws you are using to build your Robi are very small, so to help keep hold of these during each assembly stage, you may find it useful to use a magnetic screw dish. These will hold your screws securely as you go through the assembly steps, and are available online or at hobby shops.



Sometimes called a magnetic screw or hobby tray, screw dishes are available online or in hobby shops.

Testing the AA batteries

The test board and neck stand you have been using to test Robi's servos runs on AA batteries. If these run out, the board will not work, and to warn you that the batteries are running out, the LEDs will all flash rapidly (like when a servo error is experienced). A charge tester like the one pictured will help confirm this, and the solution is simply to replace the batteries with fresh AA ones.



STAGE 38: ROBI'S BATTERY PACK

With this stage you have received the battery pack that will power Robi on completion. For now, there are no assembly steps, but it is a good idea to familiarise yourself with the battery pack.

Until the battery pack is enclosed within Robi's body, it is vulnerable to damage, and as such, must be handled with care. Read the safety and care instructions in

this stage carefully, and remember especially to store the battery pack as directed, out of direct sunlight and away from high temperatures.



1



YOUR PARTS

1 Battery pack

SAFE STORAGE

When you remove the battery from its packaging, do so carefully, and read the safety notice closely before storing the pack away.



CHECKING THE BATTERY



1

Carefully unpack the battery from its protective bubble-wrap and check that it looks like the one pictured. Read the accompanying safety notice closely, then put the battery back in its protective bag and store in a cool, dry place until needed. More instructions on the battery's use and storage found below.

PRECAUTIONS AND WARNINGS

DO NOT DISASSEMBLE THE BATTERY PACK

The battery pack is tightly enclosed in a plastic wrapping; this must remain intact or there is a risk of irreparable damage.

DO NOT EXPOSE BATTERY TO LIQUID

The battery must not get wet, or be stored in damp conditions. This will cause the battery to fail and can result in electric shocks.

DO NOT EXPOSE THE BATTERY TO HEAT OR FLAMES

The battery could emit harmful gases or even explode if it is exposed to high levels of heat.

HANDLE WITH CARE

The battery is delicate: keep it away from sharp objects and be careful not to drop it at any point.

ONLY USE THE SPECIFIED CHARGER

The battery is designed only to be charged using the specified charger, supplied in the next pack.

DO NOT USE OR STORE AT HIGH TEMPERATURE

Whether in storage or when fitted into Robi on completion, extreme heat – such as radiators or a car dashboard in hot weather – must be avoided.

DO NOT SHORT-CIRCUIT THE TERMINALS

Contact with metallic items – jewellery or screws, for example – can short the battery's circuit, causing irreparable damage.

AVOID KNOCKS OR SCRAPES

Do not drop or throw the battery, and ensure its plastic sheathing remains intact.

INSTRUCTIONS FOR USE

Read the following carefully before using the battery for the first time!

- The battery provided with *Build Your Own Robi* is intended for use only with Robi, and must not be used in any other devices.
- Only use the charger provided with the series with this battery.
- Keep out of reach of children.
- If you are not going to be using Robi for a long time, it is recommended that you remove the battery.
- When the LED on the charger lights up GREEN, Robi is fully charged. Disconnect the battery from the charger immediately.
- Do not charge the battery unless it is fully drained, as this can cause overheating.
- Never leave the battery charging when unattended, such as overnight.
- Turn Robi OFF before charging, and do not turn on while connected to the charger.
- To connect and disconnect the battery, use the plastic connector – never pull on the cable itself.
- Never touch the battery with wet hands as this can cause an electric shock.
- Handle the battery with extreme care.
- Keep the plastic sheathing intact at all times.
- If you witness any odour or heat emanating from the battery, or see any physical deformation, stop using the battery immediately and do not use again.
- If you find the time between needing to recharge Robi's battery is decreasing, the battery is coming to the end of its useful life, and a replacement will be needed.
- Always dispose of batteries in accordance with local legislation.

COMING IN PACK 11

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 39

Robi's charger

PARTS PROVIDED

The mains adapter you will use to charge Robi.
(The one you receive will be suitable to your region.)

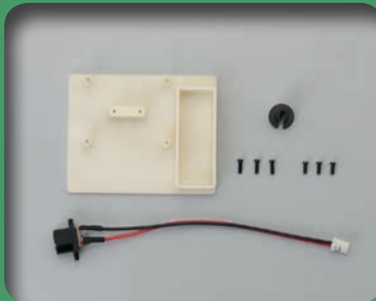


Stage 40

The power distribution board and rear hip panel

PARTS PROVIDED

The board that will distribute power to Robi's various components and the rear hip panel.



Stage 41

The power cable and CPU board support

PARTS PROVIDED

The frame that will support Robi's CPU board on completion, along with the power cord.



Stage 42

The movement test SD card

PARTS PROVIDED

The SD card you will use to test Robi's movement.